

Maths UC02 Percentages Flashcards 百分比閃卡

Print double-sided, flip on short edge. Blue border = question side · Green border = answer side. Cut along the borders. Test: 13 Oct 2026, 40 min, no calculator.

Convert: $\frac{5}{8}$

把 $\frac{5}{8}$ 化為小數和百分比

Card 1 — question

Convert: $\frac{76}{100}$

把 $\frac{76}{100}$ 化為小數和百分比

Card 2 — question

Convert: $1\frac{1}{5}$

把 $1\frac{1}{5}$ 化為小數和百分比

Card 3 — question

Convert: 0.42

把 0.42 化為分數（最簡）和百分比

Card 4 — question

Express $\frac{13}{20}$ as a percentage

把 $\frac{13}{20}$ 化為百分比

Card 5 — question

What is 0.71?

0.71 是什麼意思？

Card 6 — question

Order smallest → largest:

$\frac{5}{7}$, 0.64, 71%, 0.71

由小到大排列

Card 7 — question

Order smallest → largest:

$\frac{1}{1000}$, $\frac{11}{1000}$, 0.11%, 0.0108

由小到大排列

Card 8 — question

Convert: 5/8

$$5 \div 8 = 0.625 \rightarrow 62.5\%$$

$$\text{Chain: } 5/8 \rightarrow 0.625 \rightarrow 62.5\%$$

Card 1 — answer

Convert: 76/100

Simplify first 先約簡:

$$76/100 \rightarrow 19/25 \rightarrow 0.76 \rightarrow 76\%$$

Card 2 — answer

Convert: 1 1/5

$$1 \frac{1}{5} \rightarrow 1.2 \rightarrow 120\%$$

Percentages **can be over 100%**

百分比可以超過 100%

Card 3 — answer

Convert: 0.42

Simplify first 先約簡:

$$0.42 \rightarrow 42/100 \rightarrow 21/50 \rightarrow 42\%$$

Card 4 — answer

Express 13/20 as a percentage

$$13 \div 20 = 0.65 \rightarrow 65\%$$

$$\text{Chain: } 13/20 \rightarrow 0.65 \rightarrow 65\%$$

Card 5 — answer

What is 0.71?

A recurring decimal 循環小數:

$$0.7\dot{1} = 0.717171\ldots$$

The dot over 7 and 1 means "71" repeats
forever.

It is **slightly bigger** than $5/7 = 0.714285\ldots$

Card 6 — answer

Order: 5/7, 0.64, 71%, 0.71

Convert to decimals first:

$$0.7142\ldots, 0.64, 0.71, 0.7171\ldots$$

$$\rightarrow 0.64 < 71\% < 5/7 < 0.7\dot{1}$$

小心: 0.71 比 5/7 稍大!

Card 7 — answer

Order: 1/1000, 11/1000, 0.11%, 0.0108

Convert to decimals first:

$$0.001, 0.011, 0.0011, 0.0108$$

$$\rightarrow 1/1000 < 0.11\% < 0.0108 < 11/1000$$

陷阱: $0.11\% = 0.0011$, 比 0.0108 還小!

Card 8 — answer

Find: 17.5% of 44

求 44 的 17.5%

Card 9 — question

Find: 130% of 45

求 45 的 130%

Card 10 — question

Increase formula 加法公式

百分比增加怎麼算？

Card 11 — question

Decrease — method 1

百分比減少：方法一

Card 12 — question

Decrease — method 2

百分比減少：方法二

Card 13 — question

"X as a % of Y"

X 佔 Y 的百分之幾？

Card 14 — question

Worked: chair \$200, +15%

椅子 \$200，加價 15%

Card 15 — question

Worked: 25 questions, 20% wrong

25 題，20% 答錯，答對幾題？

Card 16 — question

Find: 17.5% of 44

Decimal multiplier:

$$44 \times 17.5\% = 44 \times 0.175 = 7.7$$

把 % 變成小數再乘。

Card 9 — answer

Find: 130% of 45

$$45 \times 130\% = 45 \times 1.3 = 58.5$$

Over 100% is fine — **just multiply!**

超過 100% 照乘。

Card 10 — answer

Increase formula

Multiply by (100% + rate):

$$\begin{aligned} 500 \text{ seats} + 16\% &\rightarrow 500 \times 116\% \\ &= 500 \times 1.16 = 580 \end{aligned}$$

乘以 (100% + 比率)。

Card 11 — answer

Decrease — method 1

Multiplier 乘數法: multiply by (100% – rate)

$$\begin{aligned} \text{bike } \$880, -15\% &\rightarrow 880 \times 85\% \\ &= 880 \times 0.85 = 748 \end{aligned}$$

Card 12 — answer

Decrease — method 2

Subtract the amount 先減數值:

20% of 25 wrong:

$$25 \times 0.20 = 5 \text{ (wrong)}$$

$$25 - 5 = 20 \text{ (correct)}$$

先算出減少了多少，再減去。

Card 13 — answer

"X as a % of Y"

$$(X \div Y) \times 100\%$$

15,300 story books of 17,000:

$$\begin{aligned} 15,300 \div 17,000 \times 100\% \\ = 0.9 \times 100\% = 90\% \end{aligned}$$

Card 14 — answer

Worked: chair \$200, +15%

$$\begin{aligned} 200 \times (100\% + 15\%) &= 200 \times 115\% \\ &= 200 \times 1.15 = 230 \end{aligned}$$

∴ The new price of the chair is \$230.

最後要寫 ∴ 結論句！

Card 15 — answer

Worked: 25 questions, 20% wrong

$$25 \times 20\% = 5 \text{ (wrong)}$$

$$25 - 5 = 20 \text{ (correct)}$$

∴ He got 20 questions correct.

Card 16 — answer

**Worked: \$250, 30% on books,
then \$145 on bag — left?**

兩步金錢題：還剩多少？

Card 17 — question

The $\therefore$ conclusion habit

結論句習慣

Card 18 — question

Test day checklist

考試日檢查表

Card 19 — question

No-calculator survival

不用計算機怎麼辦？

Card 20 — question

\$250, 30% on books, \$145 on bag

Step 1: $250 \times 30\% = 75$ (books)

Step 2: $250 - 75 = 175$ (left)

Step 3: $175 - 145 = 30$

$\therefore$ She has \$30 left.

Card 17 — answer

The $\therefore$ conclusion habit

Every word problem ends with a **stated conclusion**:

$\therefore$ The new price of the chair is \$230.

$\therefore$ She has \$30 left.

老師要求每條應用題都要寫結論句！

Card 18 — answer

Test day checklist

① Simplify fractions FIRST 先約簡

② Write **chains**: fraction $\rightarrow$ decimal $\rightarrow$ %

③ Ordering? $\rightarrow$ **decimals first**

④ Increase: $\times (100\% + \text{rate})$ · Decrease: $\times (100\% - \text{rate})$

⑤ Word problems: **one line per step + $\therefore$ sentence**

Card 19 — answer

No-calculator survival

40 minutes, NO calculator:

- Practise **decimal $\times$ decimal** by hand (e.g. 44×0.175)
- Practise **long division** (e.g. $5 \div 8 = 0.625$)
- Keep **5 min at the end** to re-check chains & conclusions

Card 20 — answer